IT Workshop II (Cloud Computing): Cloud Concepts and Technologies, Python for Cloud

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Service Level Agreement

- A SLA for cloud specifies the level of services that is formally defined as a par of the service contract with the cloud service provider.
- SLAs provide a level of service for each service which is specified in the form of minimum level of service guaranteed and a target level.
- SLAs contain a number of performance metrics and the corresponding service level objectives.
- Common criteria for cloud SLAs: Availability, Performance, Diaster Recovery, Problem Resolution, Security and Privacy of data.

SLA Requirements

- SLA format should clearly describe a service
- Present the level of performance of service
- Define ways by which the service parameters can be monitored
- Penalties when service requirements are not met

SLA Components

- Purpose
- Restrictions
- Validity period
- Scope
- Parties
- Service-level objectives (SLO)
- Penalties

SLA metrics for IaaS

Parameter	Description
CPU capacity	CPU speed for VM
Memory size	Cash memory size for VM
Boot time	Time for VM to be ready for use
Storage	storage size of data
Scale up	Max of VMs for one user
Scale down	Min number of VMs for one user
Scale up time	Time to increase number of VMs
Scale down time	Time to decrease number of VMs
Availability	Uptime of service in specific time

SLA metrics for PaaS

Parameter	Description
Scalability	Degree of use with large number of online users
Pay as you go billing	Charging based on resources or time of service
Servers	
Browsers	Firefox , IE xplorer , ..

SLA metrics for SaaS

Parameter	Description
Scalability	Using with individual or large organisations
Availability	Uptime of software for users in specific time
Customizability	Flexible to use with different types of users

SLA metrics for Storage as a Service

Parameter	Description
Geographic location	Availability zones in which data are stored
Scalability	Ability to increase or decrease storage space
Storage billing	How the cost of storage is calculated
Security	Cryptography for storage , authentication , authorization , ...
Privacy	How the data will be stored and transferred
Backup	How and where images of data are stored
Recovery	Ability to recover data in disasters or failures
Transferring bandwidth	The capacity of communication channels

Virtual Machine

- Virtual machine concept provides complete protection of system resources.
- Each virtual machine is isolated from all other virtual machines. However, it allows limited sharing of resources.
- Virtual machine system is a good vehicle for operating systems research and development.
- The virtual machine concept is difficult to implement due to the effort required to provide an exact duplicate to the simulated machine.

Snapshots and Migration

- **Snapshot:** freeze a copy of virtual machine.
  - Identify all pages in disk files, VM memory.
  - Use copy-on-write for any subsequent modifications.
- **Migration:** move a VM to another host.
  - Take snapshot.
  - Copy all pages of snapshot.
  - Copy modified pages.
  - Freeze virtual machine and copy VM memory.

Virtual Machine and Host Machine

- Virtualization layer can simulate any architecture.
- It is typically used for debugging specialized systems.
- The guest architecture does not even have to be real hardware.

Software Defined Networking

- Software-Defined Networking (SDN) is a networking architecture that separates the control plane from the data plane and centralizes the network controller.
- **Conventional Network Architecture:** The control plane and data plane are coupled. Control plane is the part of the network that carries the signaling and routing message traffic while the data plane is the part of the network that carries the payload data traffic.
- **SDN Architecture:** The control and data planes are decoupled and the network controller is centralized.

The End